

From one cable to a working plant network

Understanding Industrial Networking

Built one idea at a time

HOW TO READ THIS

- Start at Part 1 and go in order — each part adds one idea to the last
- Every part ends with a single sentence summarising it
- Only one acronym appears before Part 6, and it is explained when it arrives
- By Part 11 you can read a full plant drawing and explain every line
- The appendix is for looking things up, not for reading through

Worked example throughout: a hydro power plant with three generating units, a main and backup switch pair, control panels on fibre, and a GPS time source.

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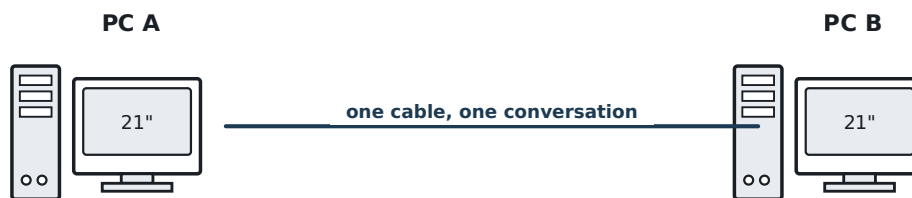
PART 1

Two Devices and a Cable

Everything in this document is built from this one picture. Each section that follows adds exactly one idea to it.

The simplest network

Two devices, one cable. Each has a **MAC address** burned into its network card at the factory — a permanent serial number like `00:1B:44:11:3A:B7`. When PC A sends data, it writes its own MAC as the sender and PC B's MAC as the recipient, wraps the data around them, and pushes the result down the wire.



works for two devices, and only two

Figure 1. Two devices connected directly. This works, and it does not scale past two.

That bundle — sender, recipient, data — is called a **frame**. It is the unit that travels on a cable. Everything else in networking is about getting frames to the right place when there are more than two devices involved.

THE ONE IDEA FROM THIS PART

Every device has a permanent hardware address. Data travels in frames that carry a sender address and a recipient address.

PART 2

Why You Need a Switch

Add a third device and the direct-cable approach collapses immediately.

The problem with cables between everything

Three devices need three cables. Five devices need ten. Twenty devices need 190. Every device would also need nineteen network ports. This is not a practical arrangement, and it is the reason switches exist.

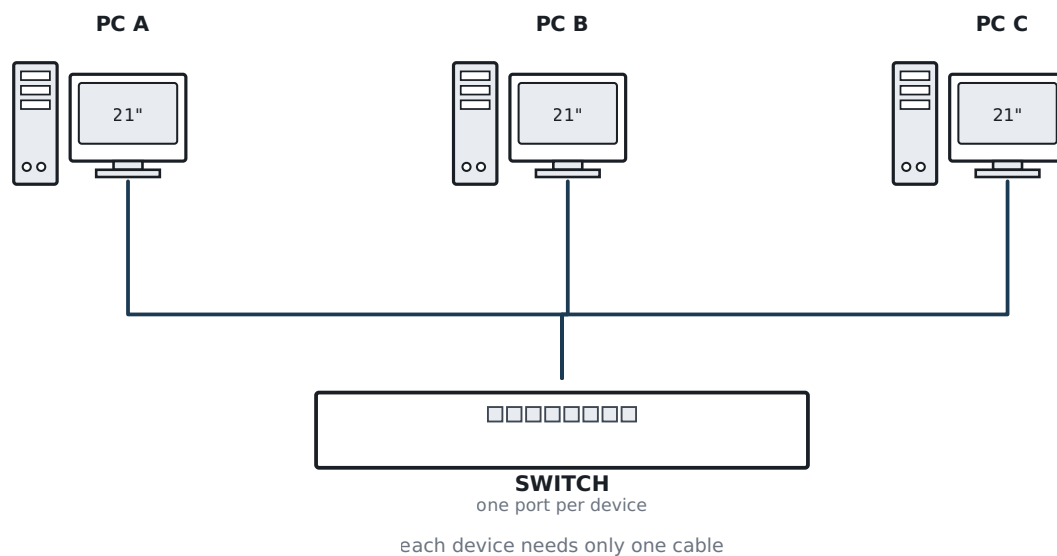


Figure 2. A switch in the middle. Every device runs one cable to it, and the switch handles who talks to whom.

Now each device has one cable and one port. Adding a device costs one cable. This shape — everything connected to one central device — is called a **star**, and it is the foundation of every network you will build.

What the switch actually does

A switch is not a splitter. It learns. Every time a frame arrives, it notes which port it came from and which MAC address sent it, and writes that pair into a table.

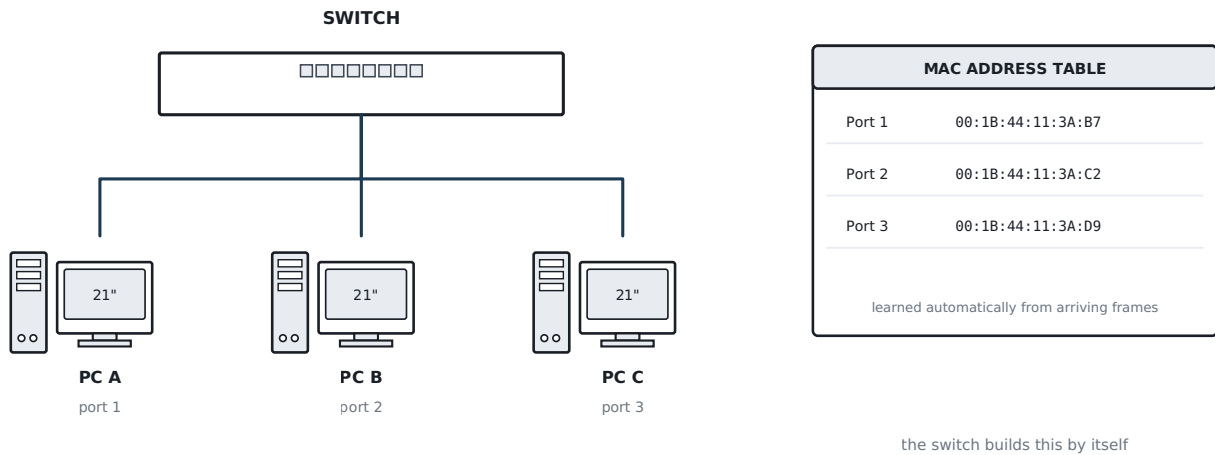


Figure 3. The MAC address table. The switch fills this in on its own by watching arriving frames — nobody configures it.

After a few seconds the switch knows where everything lives. When a frame arrives for `00:1B:44:11:3A:C2`, it looks up that address, finds port 2, and sends the frame out port 2 only. The other devices never see it.

This is the whole difference between a switch and the older device it replaced, the **hub**. A hub copied every frame to every port and had no idea what a MAC address was. A switch sends each frame to exactly one port.

THE ONE IDEA FROM THIS PART

A switch learns which device is on which port, then delivers each frame only where it needs to go.

PART 3

The Frame That Goes Everywhere

There is one exception to "delivers only where it needs to go", and that exception causes most of the trouble in the rest of this document.

Broadcasts

Sometimes a device does not know who it needs to talk to. A PC that knows the IP address `172.16.30.15` but not the MAC address behind it has to ask the whole network: *who has this address?*

It does that with a **broadcast** — a frame addressed to everyone. The switch has no choice but to copy it out of every port.

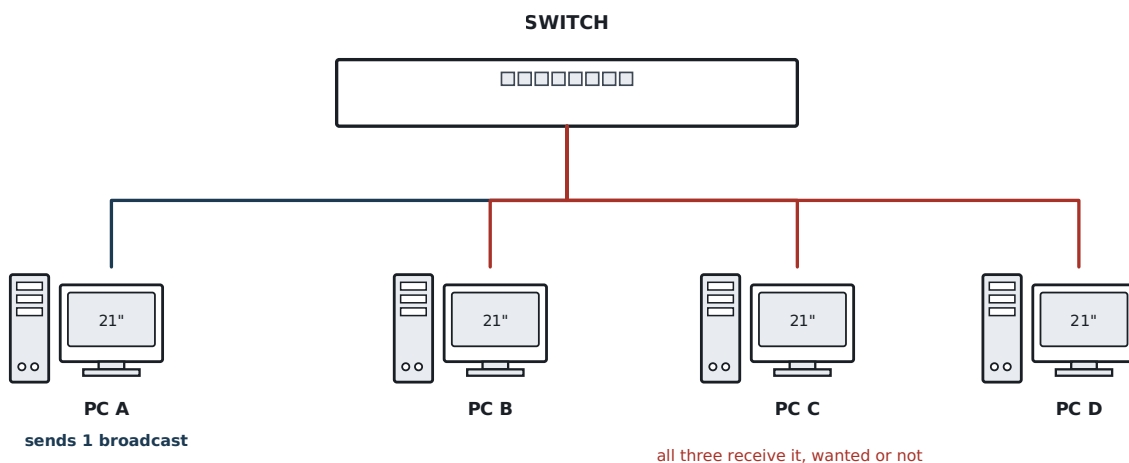


Figure 4. A broadcast. One device asks a question and every other device on the switch receives it.

Broadcasts are normal and necessary. A network with none would not function. But they scale badly: the more devices connected together, the more broadcasts everyone has to process. A group of devices that all receive each other's broadcasts is called a **broadcast domain**, and one switch is one broadcast domain.

Remember this term. Part 6 is entirely about splitting it up.

THE ONE IDEA FROM THIS PART

Some frames go to everyone. The set of devices that receive each other's broadcasts is a broadcast domain, and it grows as you add devices.

PART 4

What Happens When the Switch Dies

The star works well until the thing in the middle stops working.

A single point of failure

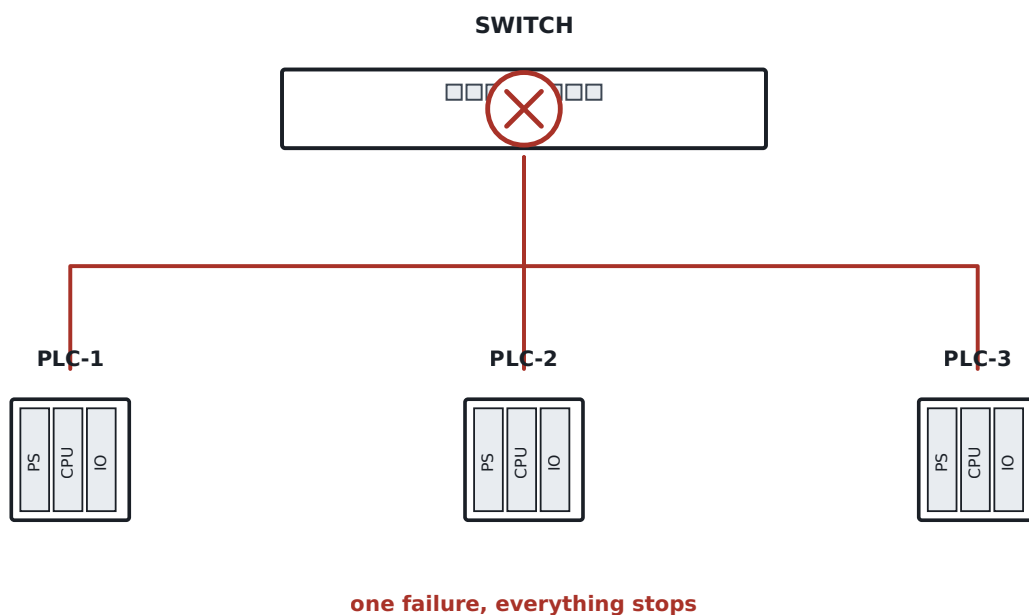


Figure 5. One switch, one power supply, one chassis. Everything below it depends on all three.

This is fine for an office. It is not fine for a plant, where the PLCs on those ports are running machinery. The obvious fix is to add a second path — a second cable, or a second switch — so that losing one thing does not stop everything.

The obvious fix creates a new and much worse problem, which is the subject of the next part.

THE ONE IDEA FROM THIS PART

A star has no redundancy. Anything with one path has one failure that stops it.

PART 5

The Loop Problem

This is the most important section in the document. Almost every rule about industrial network design exists because of what is described here.

Two cables between two switches

It seems sensible: run a second cable so that if one is damaged, the other carries on. What actually happens is that the network destroys itself in about three seconds.



Figure 6. A loop. One broadcast frame is copied back and forth between the two switches, doubling every lap, until the links are completely full.

Follow one broadcast frame. Switch 1 receives it and, because it is a broadcast, copies it out of every port — including *both* cables to Switch 2. Switch 2 now has two copies. It also floods them out of every port, so both come back to Switch 1. Now there are four. Then eight. Then sixteen.

Why it never stops

An Ethernet frame has **no expiry counter**. IP packets have one — a number that decreases at every hop and kills the packet when it reaches zero. Ethernet frames have nothing equivalent. A frame circulating in a loop circulates until someone unplugs a cable or powers off a switch.

Three things happen at once, and all three are severe:

- **The links fill up.** Real traffic cannot get through. Everything on the network stops responding.
- **The MAC tables break.** The same sender keeps arriving on different ports, so the switches rewrite their tables continuously and can no longer deliver even ordinary traffic.
- **It does not recover on its own.** Removing the cause is a physical act.

THIS IS THE ACCIDENT, NOT THE THEORY

The usual cause is not a design error. It is a technician tidying a panel who finds two free ports and a spare patch lead, and connects them. On unmanaged switches that single action takes down the plant network, and the cause is almost impossible to find by looking at the logs — because the logs cannot be reached either.

Spanning Tree: the fix

Managed switches solve this with a protocol called **Spanning Tree Protocol**, or STP. It is the only acronym you need for the first half of this document.

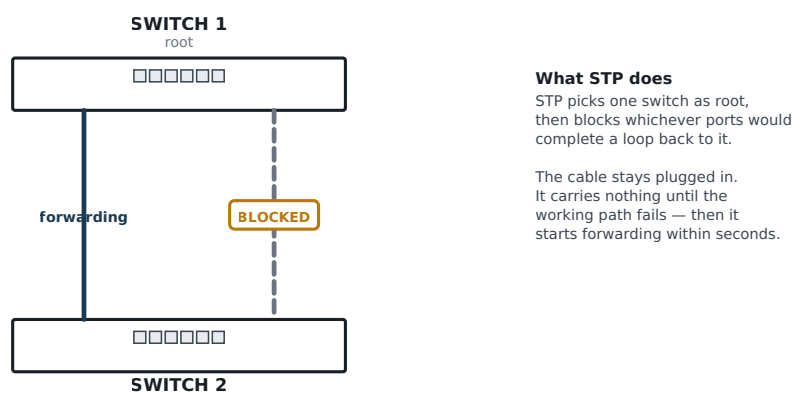


Figure 7. STP in action. Both cables are plugged in; one is deliberately switched off until it is needed.

STP works in two steps. First, the switches elect one of themselves as the **root** — a reference point. Second, each switch works out which of its ports would create a loop back to the root, and puts those ports into a **blocked** state. A blocked port has link, shows a light, and carries no data.

When the working cable fails, the blocked port notices within a second or two and starts forwarding. You get the redundancy you wanted, without the loop.

TWO THINGS WORTH KNOWING NOW

Set the root deliberately. Left alone, the switches elect whichever one has the lowest hardware address — usually the oldest switch in the least convenient cabinet. One command fixes this, and it belongs on your main switch.

Write down which port is blocked. A healthy network and a network already running on its backup path look identical from the outside. The only way to tell them apart is a record of how it should look.

THE ONE IDEA FROM THIS PART

A second path is essential and dangerous. STP makes it safe by keeping one path switched off until it is needed.

PART 6

Splitting One Network Into Several

Back to broadcast domains. Once a network has more than a few dozen devices, having them all hear each other becomes a problem — for performance, and more importantly for separation.

The case for splitting

In a plant, an operator workstation and a PLC controlling a machine have nothing to say to each other most of the time. Keeping them in the same broadcast domain means:

- The PLC spends processing time discarding traffic it never wanted.
- A misconfigured device in one area can disturb every other area.
- Anyone who plugs into any port can reach anything.

VLANs

A **VLAN** divides one physical switch into several independent switches. Ports assigned to VLAN 10 behave as if they are on one switch; ports assigned to VLAN 30 behave as if they are on a different one. They share the hardware and nothing else.

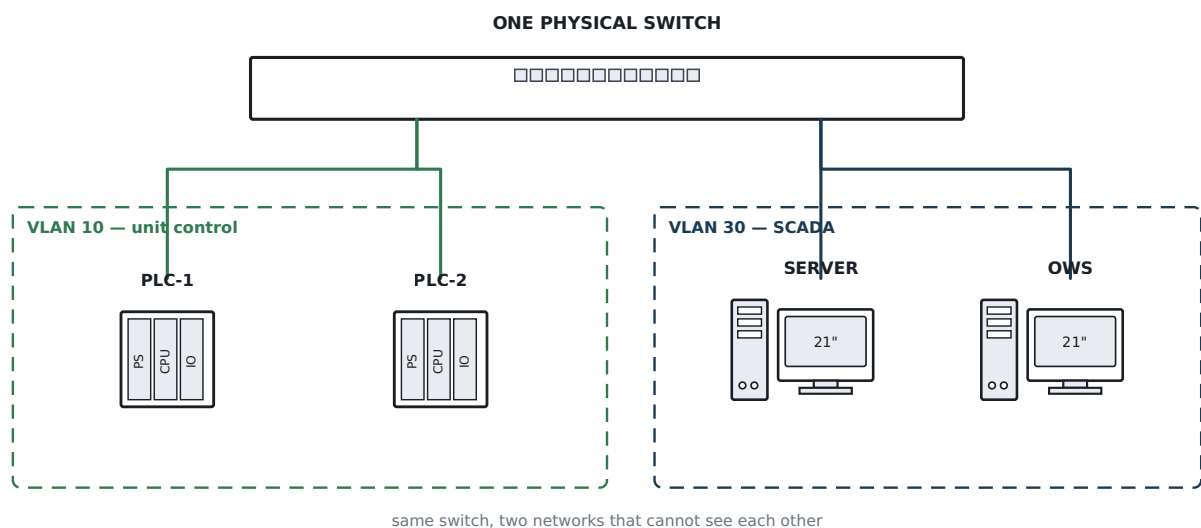


Figure 8. Two VLANs on one switch. Broadcasts in VLAN 10 never reach VLAN 30, and vice versa.

Each VLAN is its own broadcast domain. This is the tool for the problem in Part 3.

Two kinds of port

Once VLANs exist, every port on every switch is one of two kinds:

Kind	Connects to	Carries
Access	A PLC, PC, printer, HMI	One VLAN. The device has no idea VLANs exist.
Trunk	Another switch	Many VLANs, with a tag added to each frame saying which one it belongs to.



Figure 9. A trunk between two switches. One cable carries both VLANs, and a tag on each frame keeps them separate.

This is why one cable between switches is enough no matter how many VLANs you have. The tag is added when the frame enters the trunk and removed when it leaves, so the end devices never see it.

THE ONE IDEA FROM THIS PART

A VLAN splits one switch into several separate networks. Access ports belong to one VLAN; trunk ports carry many.

PART 7

Letting VLANs Talk Again

Having separated the network, you usually need controlled traffic between the pieces. That is a different job, done by a different kind of device.

IP addresses

MAC addresses have no structure — they are just serial numbers, and they only work inside one broadcast domain. To move between VLANs you need an address with structure, which is what an **IP address** provides.

172.16.30.15 with mask 255.255.255.0 means: network 172.16.30, device number 15. Everything sharing the first three numbers is local. Anything else has to be sent somewhere that knows how to reach it — the **gateway**.

Layer 2 and Layer 3 switches

	Layer 2 switch	Layer 3 switch
Forwards using	MAC address	MAC address and IP address
VLANs	Yes	Yes
Traffic between VLANs	No	Yes
Needs an IP	Only for management	One per VLAN it serves
In your plant	Panel switches	Main and backup switches

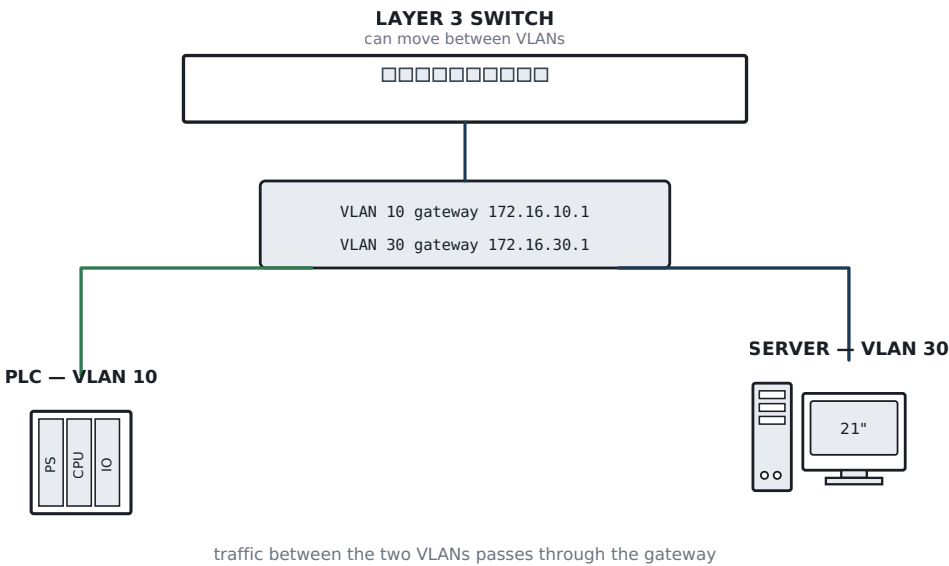


Figure 10. A Layer 3 switch holding a gateway address in each VLAN. Traffic between VLANs passes through it, and can be filtered there.

The Layer 3 switch holds one address inside each VLAN. Devices are configured to use that address as their gateway. Anything destined outside the local network is sent there, and the switch passes it into the right VLAN.

THE COMMAND PEOPLE FORGET

A Layer 3 switch does not route until you tell it to. On most platforms the command is literally `ip routing`. Without it, every address is configured correctly, every interface shows up, and no traffic passes between VLANs. It is the most common cause of "everything looks right and nothing works".

THE ONE IDEA FROM THIS PART

VLANs cannot talk to each other without help. A Layer 3 switch provides a gateway address inside each VLAN and moves traffic between them.

PART 8

Building In Redundancy

Now you have everything needed to understand why your plant is wired the way it is.

From one switch to two

Part 4 showed the problem: one switch, one failure, everything stops. Part 5 showed that adding a second path is safe as long as STP is running. Put those together and you get the arrangement in your drawing.

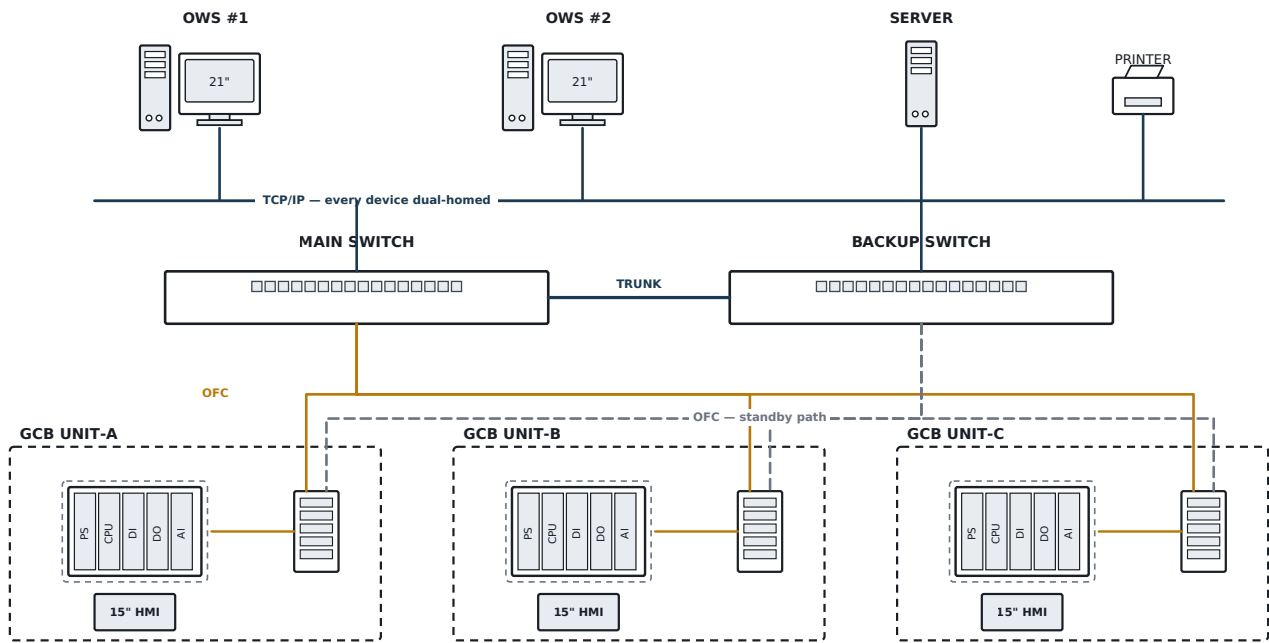


Figure 11. Dual star. Each panel has two fibre drops, one to each switch. STP keeps one blocked until it is needed.

Every panel has two cables to two different switches. That is the whole idea. Losing a cable, a switch port, or an entire switch leaves every panel still connected.

What survives what

Failure	Effect on the plant	How it recovers
One fibre drop	None	STP unblocks the other, 1-3 seconds
Main switch	None	All panels move to backup
Backup switch	None	Main was already carrying traffic
One panel switch	That panel only	Manual replacement
Both switches	Everything	Manual — see the warning below

THE MISTAKE THAT MAKES REDUNDANCY FAKE

Two switches in the same cabinet, on the same DC supply, fed from the same breaker, are not redundant. They are one switch that costs twice as much. Separate power feeds, separate cable routes, and ideally separate cabinets — otherwise a single fuse takes out both halves of a design that looks fully redundant on the drawing.

Other shapes you will hear about

Your plant uses a dual star. Three other arrangements exist, and this table is all you need unless you are asked to build one:

Shape	How it is wired	Recovery	Used when
Dual star	Every panel to two switches	1–3 s	Your plant
Ring	Panels chained, last joined to first	0.03–3 s	Cable follows the machine hall
Dual ring	Two rings joined by a link	same	Separating two traffic types
PRP / HSR	Every frame sent twice on two paths	0 s	Electrical protection, where no loss is allowed

They all solve the same problem — surviving a broken cable — and differ only in how fast and how expensively. Your dual star recovers in a second or two, which is comfortable for the poll rates in a hydro plant.

PART 9

Talking to Field Equipment

The network exists to move process data. This part covers what that data looks like on the wire.

Modbus TCP

The most common protocol between a SCADA system and field equipment, and the one your mixed-vendor plant uses. It is deliberately simple.

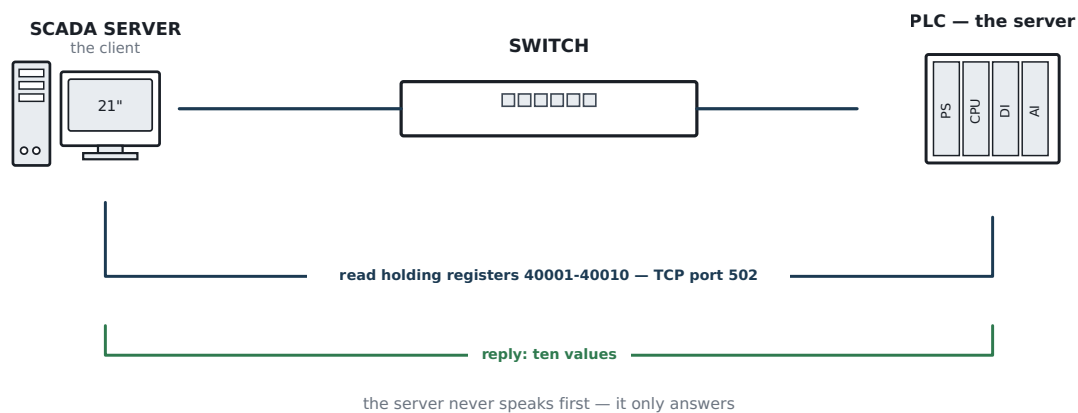


Figure 12. A Modbus TCP exchange. The client asks, the server answers, and nothing happens without a request.

Data lives in numbered registers on the device. The SCADA system opens a connection to **TCP port 502** and asks for a range of them. The device replies with the values. That repeats every poll interval, forever.

Register type	Holds	Typical use
Coil	One bit, read/write	Start command, output
Discrete input	One bit, read only	Limit switch, status
Input register	16 bits, read only	Measured value
Holding register	16 bits, read/write	Setpoint, reading

THREE THINGS THAT WILL COST YOU TIME

No security at all. Modbus has no password and no encryption. Anyone who can reach port 502 can write to it. Separation by VLAN and a rule limiting which addresses may connect is the only protection there is.

Register numbering is inconsistent. Some manuals count from zero, some from one, and the "40001" convention is applied differently by different manufacturers. Byte order for values that span two registers also varies. Check against live readings during commissioning; do not trust the manual.

Timeouts must exceed recovery time. If STP takes two seconds to switch to the backup path and your poll timeout is one second, every normal failover will mark devices as dead. Set the timeout comfortably above the worst case.

Devices that are not on Ethernet

Temperature scanners, gauge panels and older meters usually speak Modbus over a serial wire (RS-485) rather than Ethernet. A **serial gateway** sits between them and the network, appearing as a normal Modbus TCP device.

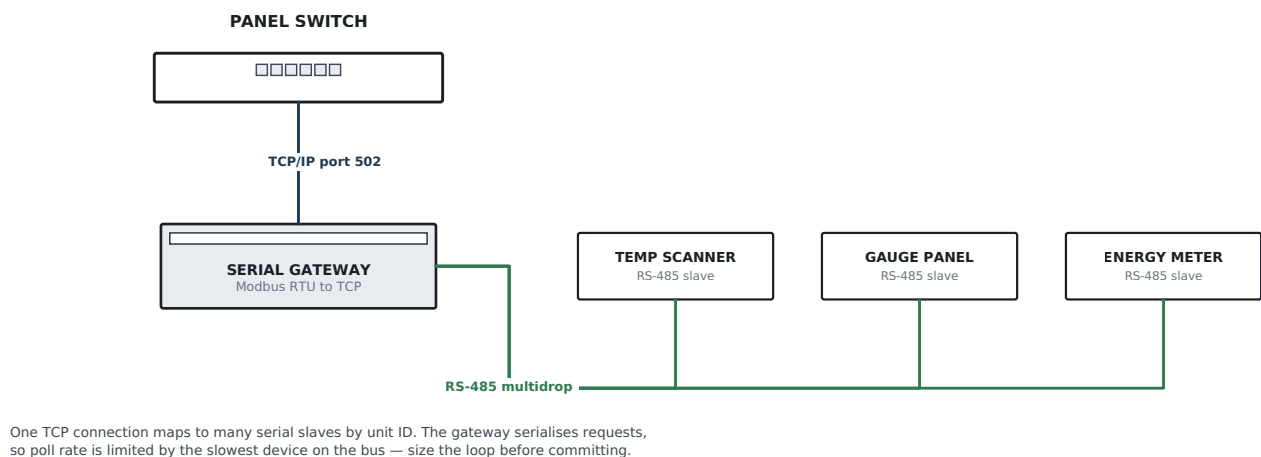


Figure 13. A serial gateway. Several RS-485 devices share one wire and appear on the network through one address.

The important consequence: the gateway can only talk to one serial device at a time. Ten devices at 100 ms each means a full cycle takes a second, no matter how fast the network is. Size this before promising a poll rate.

PART 10

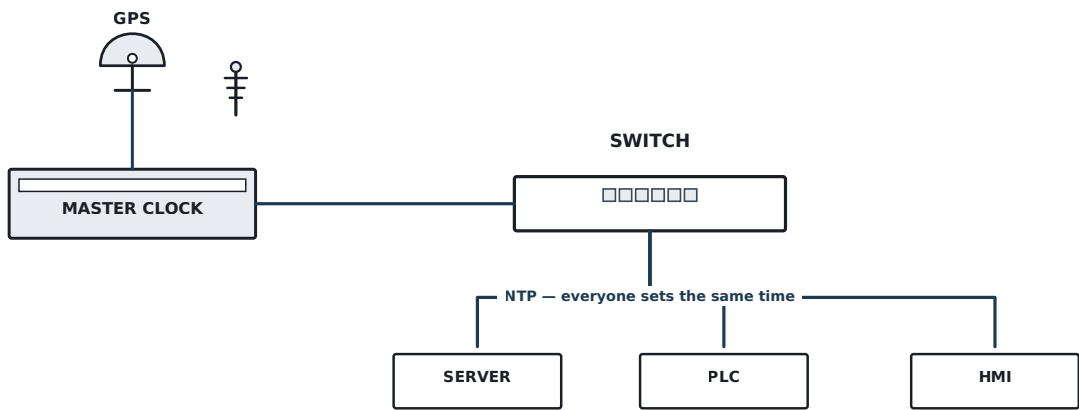
Why Everything Needs the Same Time

The GPS antenna in your drawing is not an optional extra.

The problem it solves

A trip occurs. Three devices record it: the unit controller, the switchyard panel, the SCADA server. Each writes a timestamp from its own internal clock. Those clocks drift by seconds a day, so the three records disagree about the order of events.

Now try to answer "what happened first". You cannot. The record is unusable for exactly the event it was there to capture.



without this, two devices log the same event at different times

Figure 14. A GPS clock distributing time over the network. Every device sets its clock to the same source.

How accurate is accurate enough

What you want to do	Accuracy needed	Method
Readable logs	about a second	NTP
Correlate alarms	tens of milliseconds	NTP
Order breaker operations	1 millisecond	IRIG-B wire, or PTP
Electrical fault analysis	1 microsecond	PTP with capable switches

NTP sends time over the ordinary network and gets within a few milliseconds. It covers everything most plants need. **IRIG-B** is a dedicated wire carrying a timing signal, used for devices needing better than a millisecond — it does not depend on the network at all, which makes it very robust.

THE FAILURE NOBODY NOTICES

When a GPS clock loses its satellite signal it does not stop. It keeps producing time, and slowly drifts. Nothing alarms, nothing looks wrong, and you discover it weeks later when a fault record turns out to be useless.

Every GPS clock has an alarm contact and a status you can read over the network. Wire it into SCADA. It takes five minutes and it is the difference between knowing and not knowing.

PART 11

Reading Your Own Drawing

Everything up to here was preparation for this. You should now be able to explain every symbol and every line on your plant drawing.

The symbols

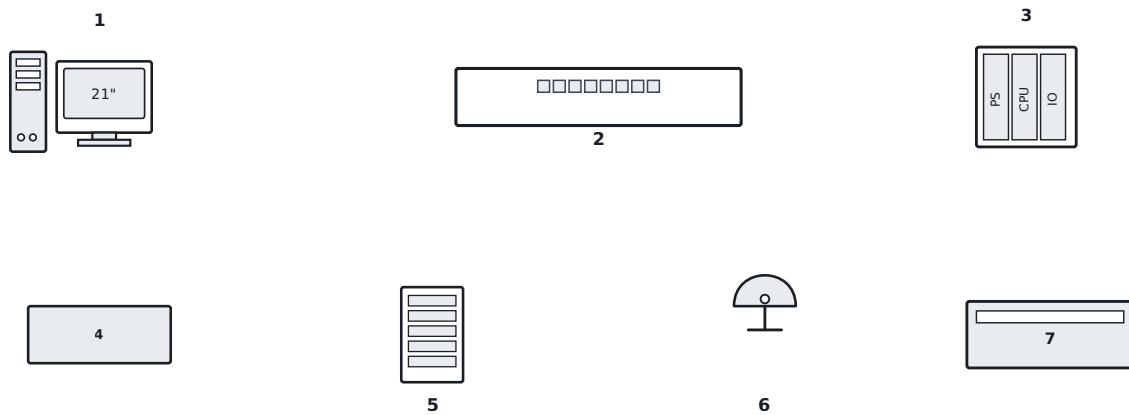


Figure 15. The symbols used on plant network drawings, with what each one means.

The whole scheme

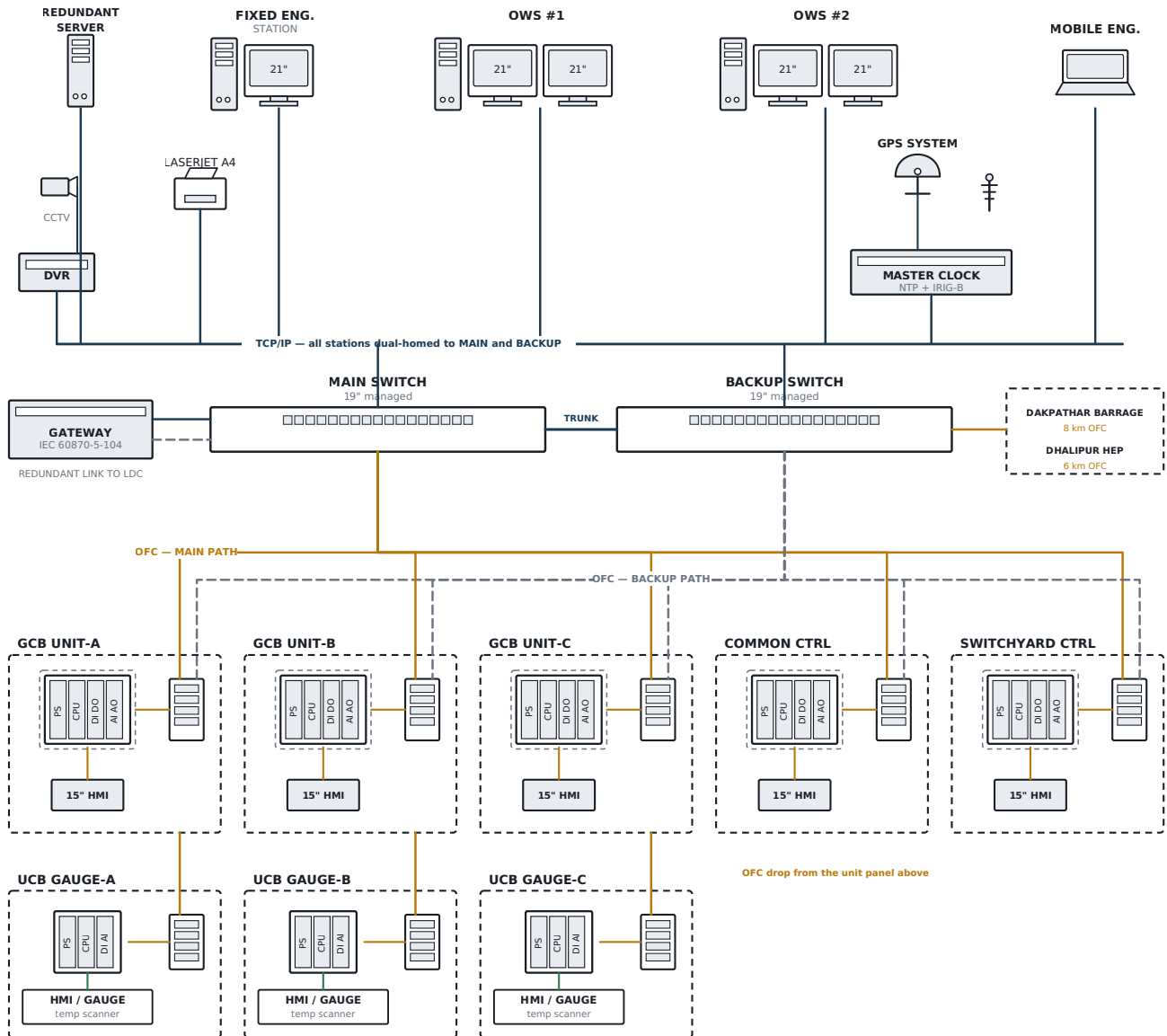


Figure 16. Your plant. Station equipment at the top, the switch pair in the middle, control panels below, remote sites to the right.

Reading it in order

- 1. Top row — the people.** Servers, operator workstations, the engineering station, printers. These are ordinary computers on copper cable.
- 2. The two switches.** Main and backup. Everything above and below connects to both. This is the dual star from Part 8.
- 3. The gateway on the left.** The only device that talks to the outside world, converting plant data into the format the load despatch centre expects. Nothing else is reachable from outside.
- 4. The GPS clock on the right.** Part 10. Feeds time to everything.

5. **Fibre down to the panels.** Two drops per panel, one from each switch. Fibre rather than copper because panels sit on different earth points, and a copper cable between them is a hazard during a fault.
6. **Each panel.** A controller (often a redundant pair), a small switch, and a local HMI so the unit can be operated even if the station network is completely down.
7. **The gauge panels below.** Same pattern, smaller, with temperature scanners on serial wire behind a gateway — Part 9.
8. **Remote sites.** Two locations several kilometres away on their own fibre.

CHECK YOUR UNDERSTANDING

If you can answer these without looking back, the document has done its job:

1. Why does each panel have two fibre cables instead of one?
2. What would happen if someone connected a spare cable between two panel switches?
3. Why is fibre used between panels but copper in the control room?
4. If the main switch is powered off, what stops working?
5. Why does the GPS clock matter for fault investigation?

APPENDIX

Reference

Terms

Term	Meaning
Frame	The unit of data on a cable, carrying sender and recipient MAC addresses
MAC address	Permanent hardware address of a network port
IP address	Configurable logical address, structured so traffic can be routed
Broadcast	A frame sent to every device in the network
Broadcast domain	The group of devices that receive each other's broadcasts
Switch	Device that learns MAC addresses and delivers frames to the right port
Managed switch	A switch you can configure — VLANs, STP, monitoring
VLAN	A logical division of one switch into several separate networks
Access port	Switch port belonging to one VLAN, connecting an end device
Trunk port	Switch port carrying several VLANs between switches
STP	Spanning Tree Protocol — blocks ports to prevent loops
Root	The switch STP uses as its reference point
Blocked port	A port with link that carries no data, held ready by STP
Gateway	The address a device sends traffic to when the destination is not local
Layer 2 / Layer 3	Forwarding by MAC address / by IP address
Modbus TCP	Simple request-and-reply protocol on TCP port 502
RS-485	Serial wiring standard used by older field devices
NTP	Time synchronisation over the network, accurate to milliseconds
IRIG-B	Time signal on a dedicated wire, accurate to microseconds
OFC	Optical fibre cable

Common faults

Symptom	Likely cause	Check
Whole network dead, all lights solid	Loop — a spare cable was connected	Disconnect suspect cables until it clears
Two VLANs cannot reach each other	ip routing not enabled	Check the routing table is not empty
Device takes 30 s to respond after reboot	STP delay on that port	Enable the fast-start option on access ports
Intermittent SCADA timeouts	Speed or duplex mismatch	Look for rising error counters on the port

Symptom	Likely cause	Check
Redundancy worked once, never again	Path did not return after repair	Check which port is blocked against the record
Modbus values wrong but connection fine	Register offset or byte order	Compare against a known live value
Timestamps disagree between devices	Clock source lost or not configured	Check every device points at the GPS clock

Practising in Packet Tracer

Concept	Possible?	Note
Star, tree, MAC learning	Yes	Use simulation mode to watch frames move
Broadcast behaviour	Yes	Send a ping and watch the flood
Loop and storm	Partly	Use hubs — switches run STP by default and will not storm
STP, root election, blocked ports	Yes	The best thing to practise here
VLANs, access and trunk ports	Yes	Full support
Inter-VLAN routing	Yes	Use a 3560 and remember <code>ip routing</code>
Dual star with two switches	Yes	Builds your plant's shape exactly
Modbus, serial gateways	No	Not simulated
NTP	Basic	Enough to see the hierarchy
PRP, HSR, IRIG-B	No	Needs real hardware